VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID

VK26-IITI-HT-0050

Dermasense: a laser-written strip that reads a dressing's own 'change line'

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Health Tech: Emergency Response & Remote Monitoring

Evidence key: • verified • designed, lab test pending • future scope

Problem Context

PUBLISHED EVIDENCE · CITED

44.9 %	≈ 2 cm	1 point	Uneven
OF 588 CHANGES MADE AT OPTIMAL MOISTURE[1]	MAKER'S 'CHANGE NOW' LINE FROM PAD EDGE[2]	WHERE TODAY'S MOISTURE SENSOR READS[1]	IN-VITRO MOISTURE MAP, 8 POINTS[3]

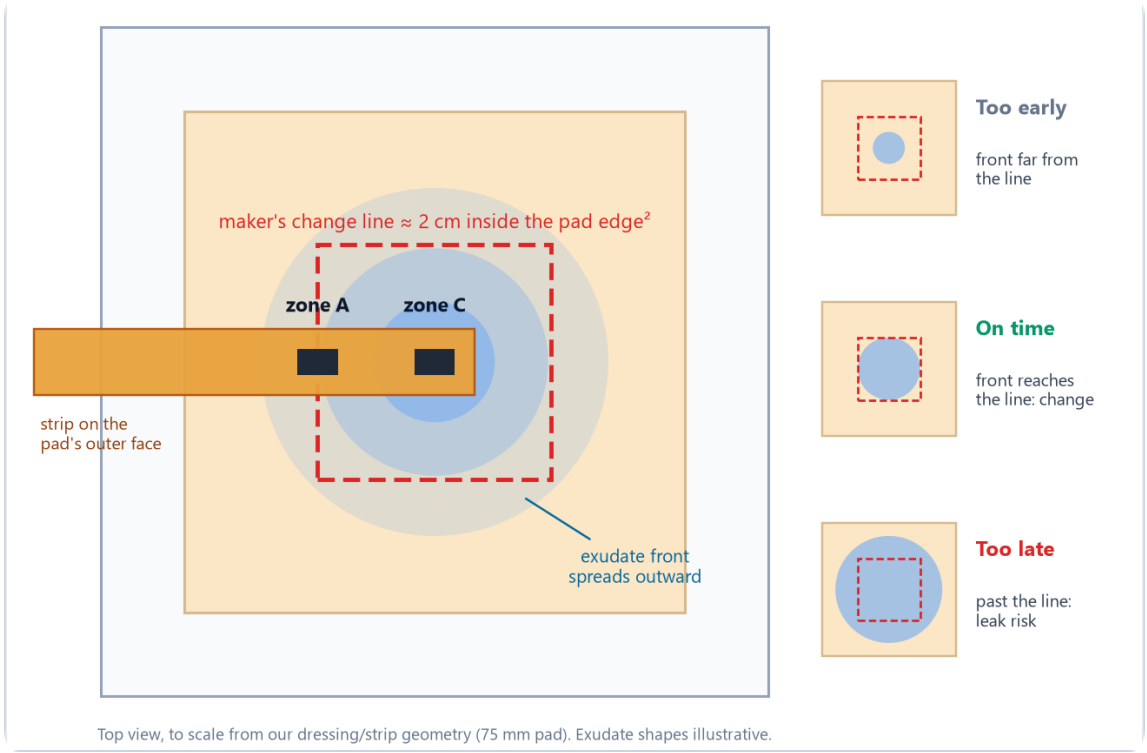


Figure: the change line is invisible under a cover (to scale, our design).

Target Beneficiary & Problem Statement

- **Who:** nurses and home carers of chronic wounds under foam dressings.
- **Pain:** the 'change now' sign is visual, and the dressing is covered.
- **Today:** change by the clock, by lifting the cover, or after a leak.
- **Gap:** in our search, no low-cost sensor reads that change line[4][5].

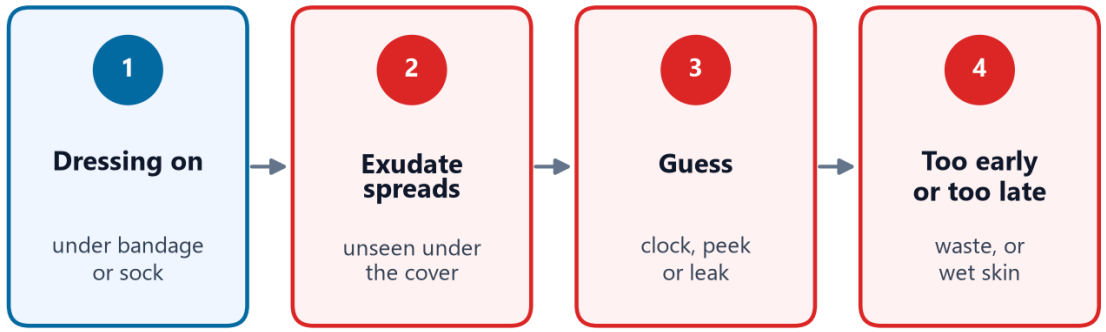
Problem: carers cannot see when a covered dressing is really full.

User Journey: Before State

CURRENT STATE · CITED

44.9 %	0.1–1.3 g/cm²·day	Hidden	Lift to look
CHANGES WHILE MOISTURE WAS STILL OPTIMAL[1]	REPORTED EXUDATE RATES, 2 METHODS[6]	EXUDATE FRONT UNDER BANDAGE OR GARMENT	THE ONLY WAY TO CHECK A COVERED PAD

Today: checked by schedule or by looking



44.9 % of changes came while moisture was still optimal[1]

Cited numbers only (References slide)

Figure: today's pain-point map (cited numbers only).

Current Pain Point Mapping

- **Time:** every check means unwrap, look, re-secure.
- **Waste:** 44.9 % of changes came while moisture was still optimal[1].
- **Risk:** a late change can leak: wet skin, soiled clothing.
- **Cause:** reported exudate rates span over $10\times^{6}$; a clock cannot follow.

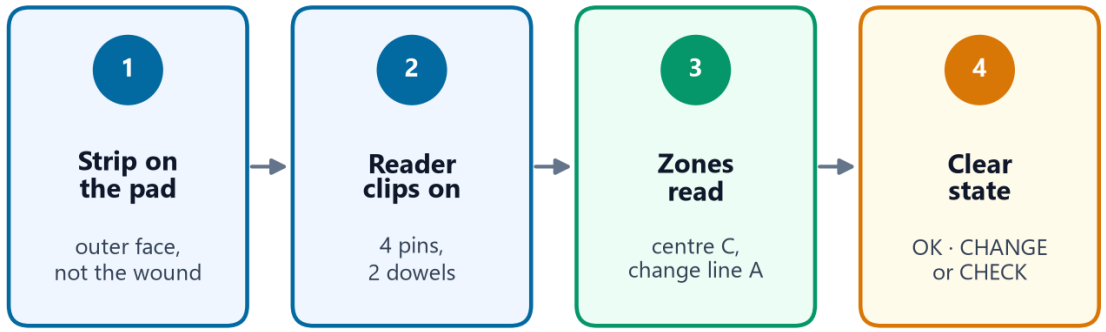
Quantified Inefficiency: published numbers only; nurse time will be measured in our interviews.

User Journey: After State (Intended)

INTENDED · NOT CLINICALLY VALIDATED

2 zones	4 states	±10 %	0
CENTRE C + MAKER'S CHANGE LINE A	OK · CHANGE · CHECK · FAULT	TARGET ALERT ACCURACY (BENCH, E4)	CONTACT WITH THE WOUND BED (BY DESIGN)

Intended: the strip reads the change line



Target: alert within ±10 % of the visible front (bench test E4)

INTENDED WORKFLOW · NOT CLINICALLY VALIDATED

Figure: the intended workflow; bench-testable, not clinically validated.

Future Solution Mapping & Quantified Efficiency

- **Same routine:** strip on the pad, reader clips on.
- **CHANGE:** when the change line is wet.
- **CHECK:** if the line wets before the centre (leak).
- **Target:** alert within ±10 % of the visible front (E4).

Design Principle: only film and carbon are disposable; the strip never touches the wound.

System Architecture & Specs

DESIGNED · LAB TEST PENDING

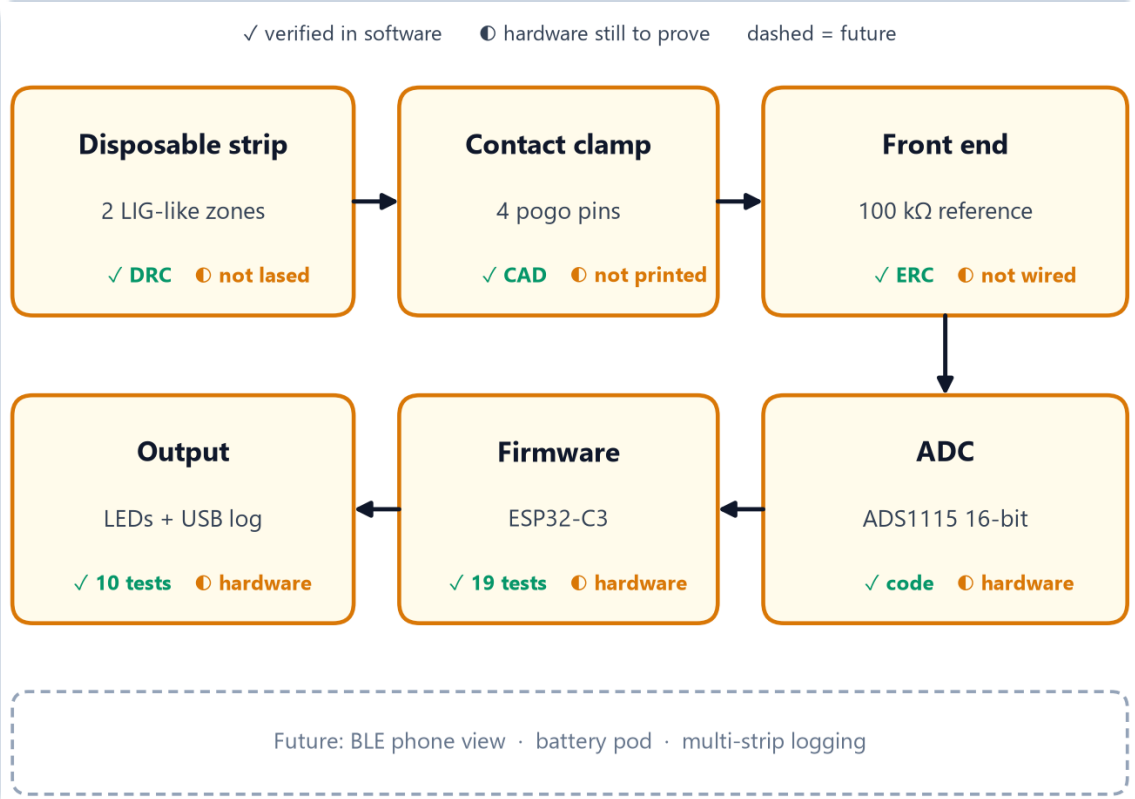


Figure: subsystems; ✓ = verified in software, ◐ = hardware still to prove.

Target Specifications & Hardware

Parameter	Design value
Sensing zones	2: pad centre C, change line A
Wet / dry	< 100 kΩ / > 300 kΩ, 3 scans
Drive	bipolar, ≤ 33 µA, no net DC
ADC / MCU	ADS1115 16-bit · ESP32-C3
Contacts	4 pogo pins + 2 dowels
Output	LEDs + USB log (BLE later)

Architecture Rationale: only film and carbon are disposable; every electronic part is reused.

System Architecture & Specs: Firmware & Data Flow

VERIFIED IN SOFTWARE · HARDWARE PENDING

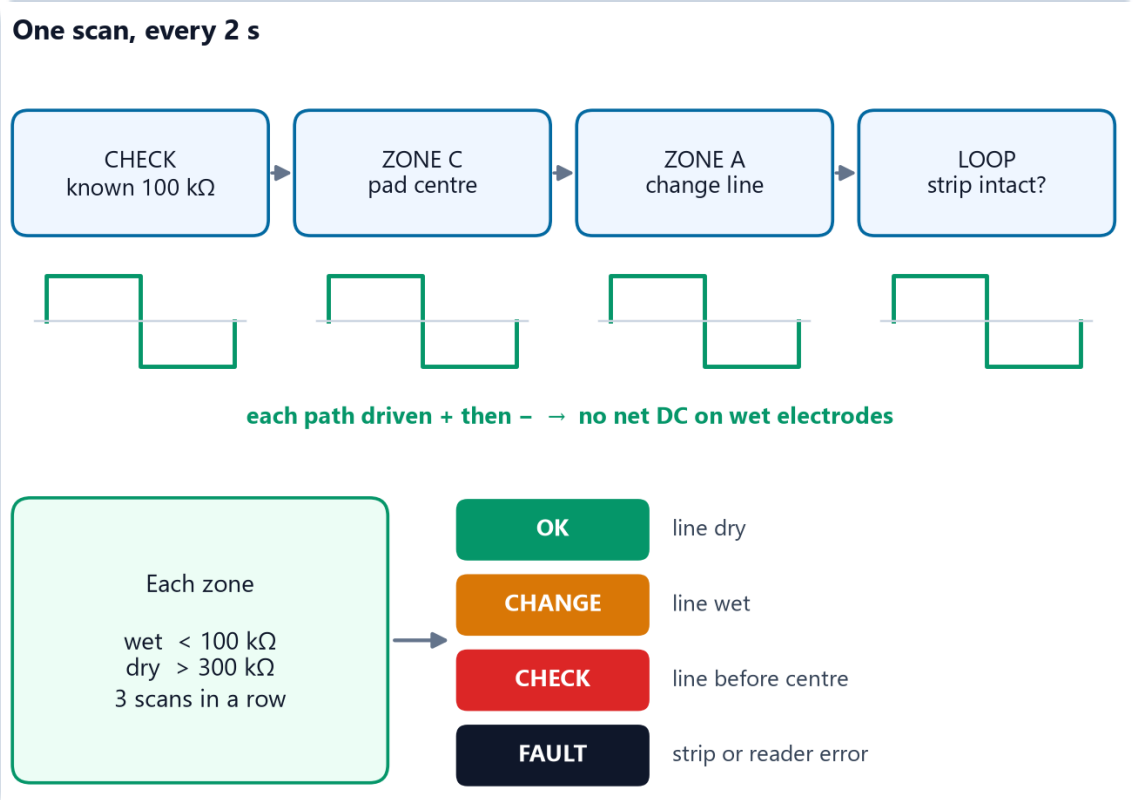


Figure: one reader scan, from measurement to state.

Firmware Logic & Data Flow

- **Measure:** each zone driven both ways, so no net DC.
- **Read:** 16-bit ADC, scaled to the 3V3 rail.
- **Decide:** wet < 100 kΩ, dry > 300 kΩ, 3 scans in a row.
- **Self-check:** a known 100 kΩ every scan, else FAULT.
- **Log:** one CSV line per scan over USB.

Software vs Hardware: 19 logic tests and 10 PC tests pass; not yet run on a wired board.

CAD Design & Mechanical Specs

DESIGNED · CAD VERIFIED · NOT YET BUILT

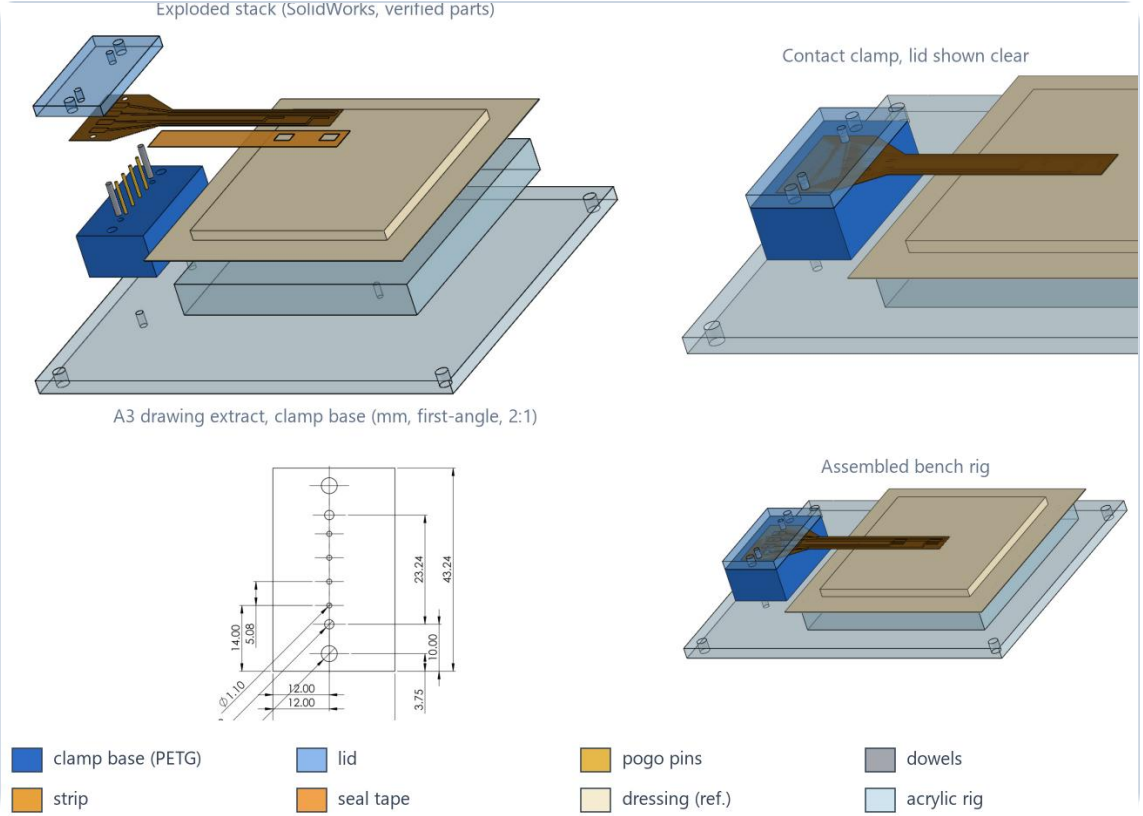


Figure: our SolidWorks 2019 models: exploded stack, clamp, drawing, assembly.

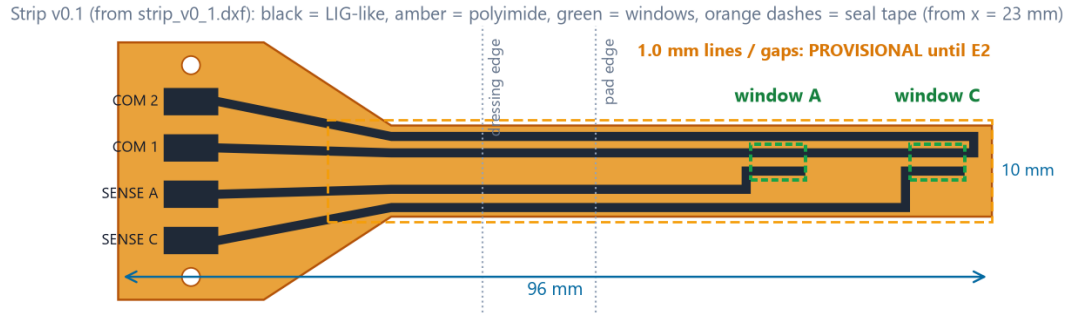
Materials & Mechanical Checks

- **Materials:** polyimide strip · PETG clamp · acrylic rig.
- **Contact:** 4 pogo pins, 0.72 N each (calculated).
- **Alignment:** 2.8× tolerance margin on 3 mm pads.
- **Verified:** 99 automated CAD checks pass, 0 fail.
- **Provisional:** strip sizes wait for E2 laser data.

Design Iteration: a clash check moved the seal tape off the clamp (decision D-013).

CAD Design & Mechanical Specs: Sensing Strip & Laser Process

DESIGNED · LAB TEST PENDING



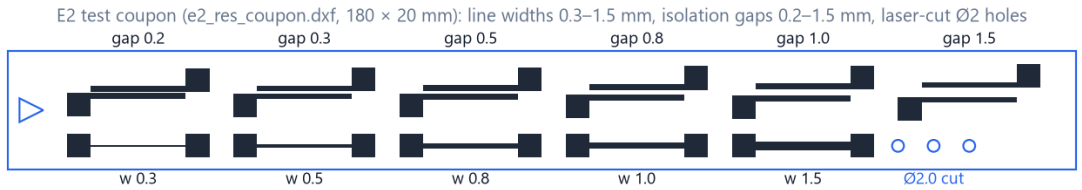


Figure: our laser files, drawn from the DXFs. Not yet lased.

Fabrication & Process Validation (E2)

- **Material:** laser-carbonised, LIG-like layer on polyimide[7].
- **Reference:** setting '12', most likely % PWM, not watts.
- **E2 plan:** 7 coupons, 2 days, 72 pre-registered samples.
- **Pass rules:** CV ≤ 20 %, gaps > 3 MΩ, fixed beforehand.
- **Sealed:** leads under tape; LIG made in air wets easily[9].

Laser Files Verified: 35/35 file checks pass; LaserCAD import is checked at the machine.

Electronics Schematics & BOM

ERC VERIFIED · NOT YET WIRED

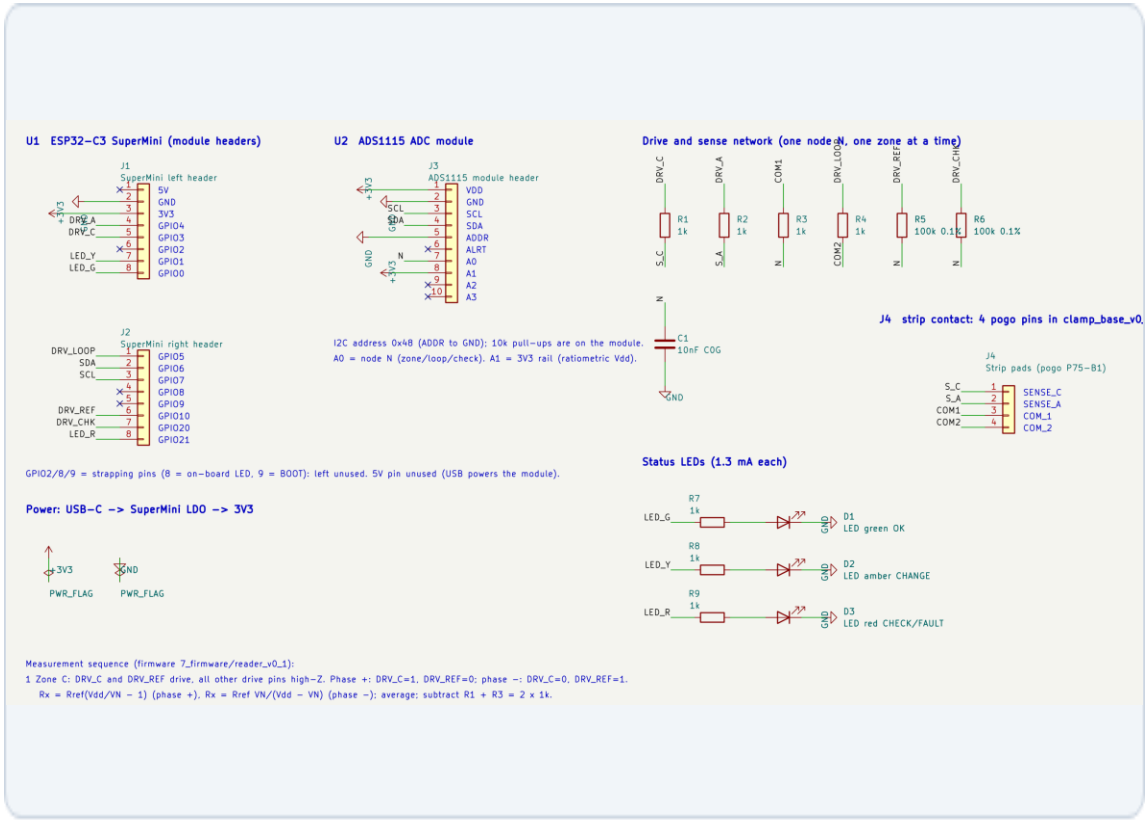


Figure: our KiCad 10 schematic of the reader (exported by kicad-cli).

Key Connections

- **Drive:** GPIO3 / GPIO4 → 1 kΩ → zones C, A.
- **Sense:** node N → 100 kΩ reference → ADC A0.
- **Check:** 100 kΩ on GPIO20; strip loop on GPIO5.
- **Strip:** 4 pogo pins on 3 × 6 mm pads.
- **Power:** USB-C → 3V3 regulator on the module.

Verification: ERC 0 violations; netlist matches firmware and strip pads (0 mismatches).

Electronics Schematics & BOM: Bill of Materials

ESTIMATES · NOT YET ORDERED

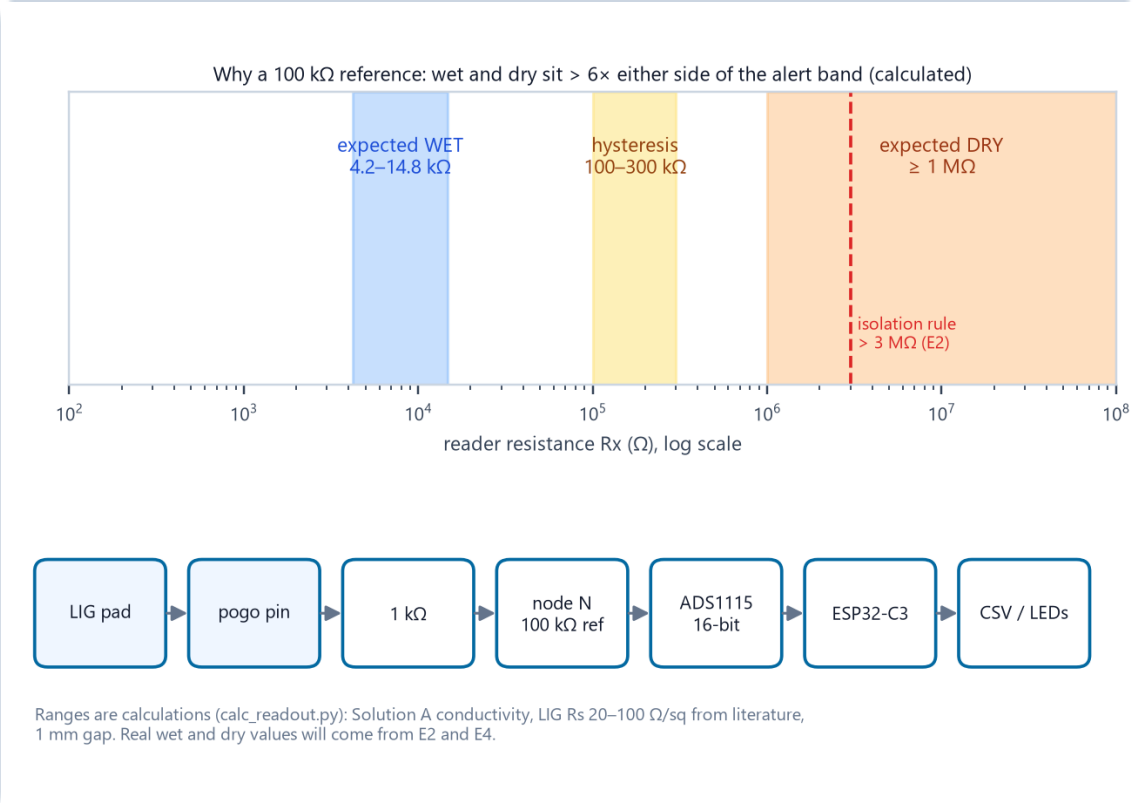


Figure: why a 100 kΩ reference (calculated ranges).

Prototype Bill of Materials

Item	Est. ₹
Reader: ESP32-C3 + ADS1115	450–900
Resistors, capacitor, LEDs, board	158–421
Pogo pins P75-B1	90–240
Clamp + bench rig	100–1,240
Strip materials	800–3,500
Test consumables	480–1,060
Total (estimate)	**≈ 2,078–7,361**

Procurement Status: nothing ordered yet; all parts sold in India; prices to confirm.

Maker Journey

GENUINE ARTIFACTS · NO STAGED EVIDENCE

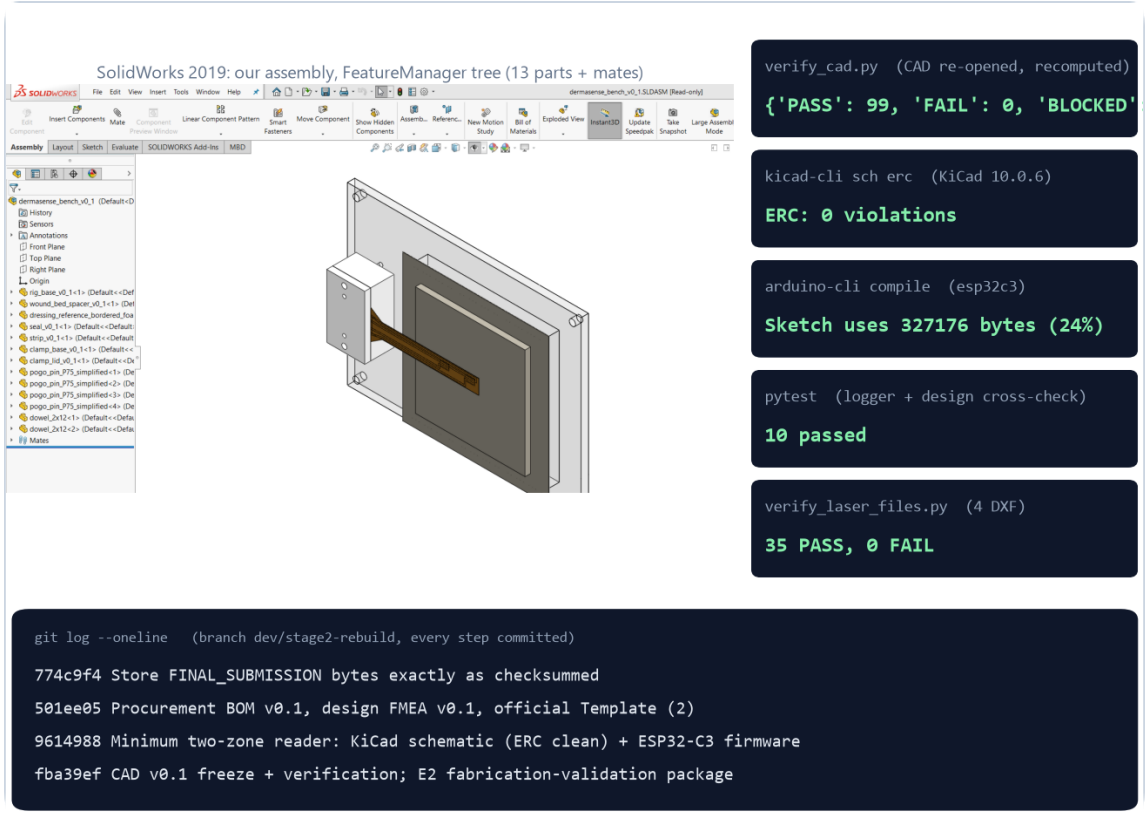


Figure: our SolidWorks window and verbatim tool output (14–15 Oct 2026).

Authenticity: What Each Artifact Proves

- **CAD:** our assembly open in SolidWorks, 13 parts.
- **Checks:** CAD, ERC, compile and tests all re-runnable.
- **History:** every step a git commit since 14 Oct.
- **Photos:** none yet; the first laser session is next.

Raw Repository Linkage: [team to insert the GitHub / Drive link before upload]

Maker Journey: Engineering Iterations

DOCUMENTED DECISIONS

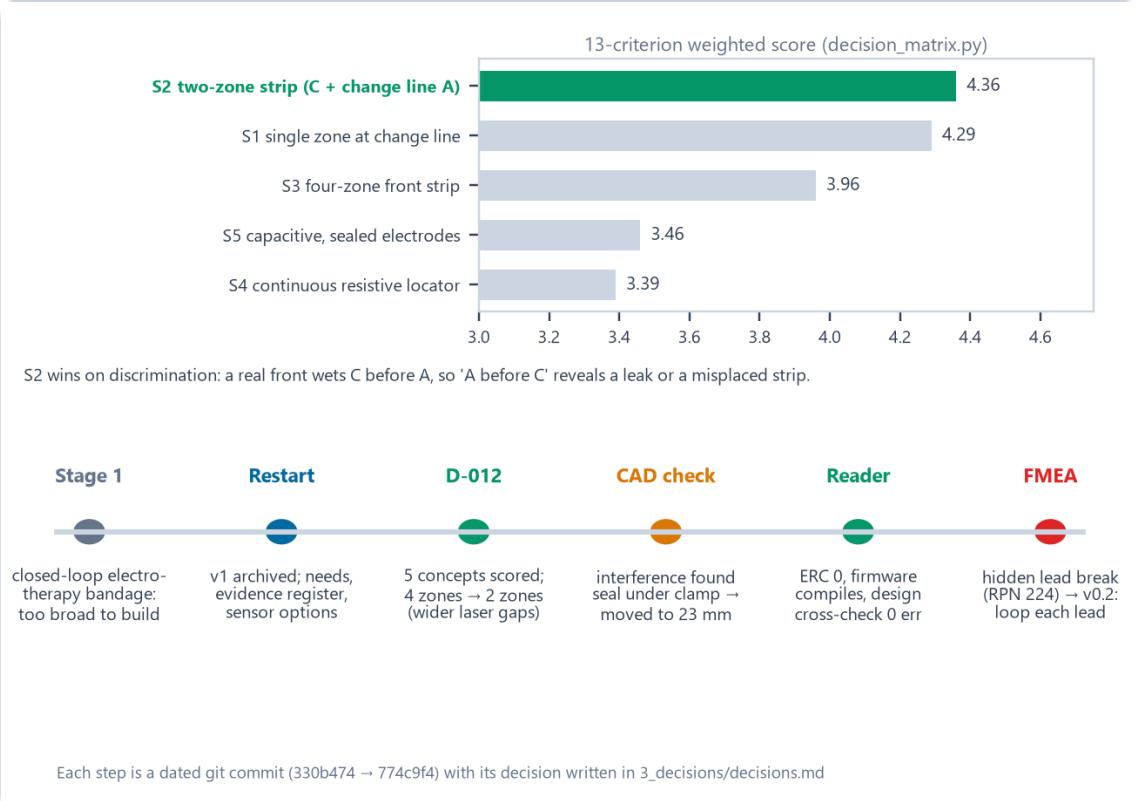


Figure: concept scores and the iteration path (each a dated commit).

Stage 1 to Stage 2 Iterations

- **Pivot:** broad electrotherapy bandage → one job: is it full?
- **Choice:** two-zone strip (4.36) beat one zone (4.29).
- **Why:** its order check tells a real front from a leak.
- **Simplify:** 4 zones → 2 until E2 proves finer lines.
- **Redesign:** FMEA risk → v0.2 loops each sense lead.

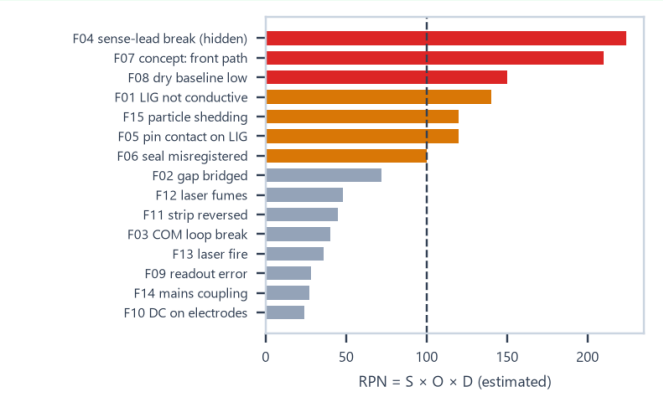
Prior Art: diaper wetness strips exist[4]; our novelty is the change-line placement and order check.

Safety & Hazard Mitigation

RISKS ESTIMATED · TESTS PLANNED

Design FMEA: top risks of 15 (ratings estimated before tests)

Failure mode	RPN	Fix
Cracked sense lead reads 'dry'	224	v0.2: loop each lead to a 2nd pad
Front path differs from model	210	test on real foam dressings (E4)
Humid dressing reads 'wet'	150	measure dry baselines (E4)



Fail-safes

- **Bench only:** USB reader, never on a person.
- **No net DC:** bipolar drive, idle pins off.
- **Self-check:** every scan, else FAULT.
- **No wound contact:** strip on the pad's outer face.

Standards (considered, not claimed)

- **ISO 14971** risk management
- **IEC 60601-1** future on-body reader
- **ISO 10993-1** before any skin contact[11]
- **EN 13726** simulated exudate for tests[10]
- **IEC 60825-1** laser work during fabrication

Roadmap

FUTURE SCOPE · GATED PLAN

4-Month Plan: each gate must pass before the next step



Build & Bench (Months 1 - 2)

- **Nov:** E2 laser tests; build and bench-test the reader.
- **Dec:** strip v0.2; print clamp; contact and bend tests.

Validate & Demo (Months 3 - 4)

- **Jan:** strips on real dressings with simulated exudate.
- **Feb:** repeatability; integrated bench demo.
- **Stage 3:** OK → CHANGE shown live on a dressing.

Technical Challenges & Risks

FOUND IN OUR OWN VERIFICATION

Geometry: Seal Under Clamp

Found: seal tape overlapped the clamp (9.0 mm³).

Impact: would lift the strip off the pins.

How: automated clash check.

Circuit: Hidden Lead Break

Found: a cracked sense lead reads 'dry'.

Impact: CHANGE never shown, no alarm.

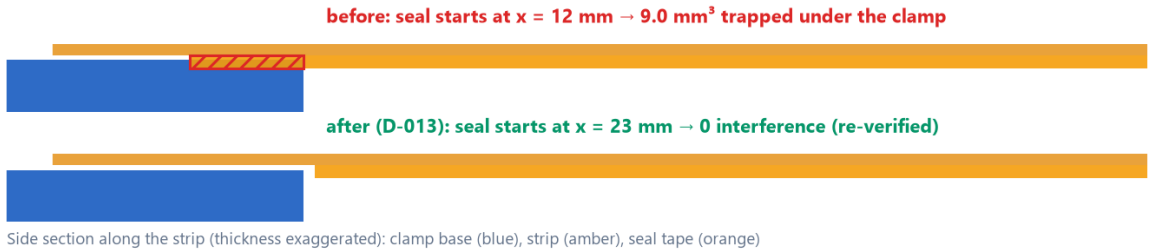
How: design FMEA (RPN 224).

Process: Unknown Laser Recipe

Found: '12' is likely a % command; tubes are unsteady at low power[8].

Impact: every strip size is provisional.

How: LaserCAD config audit.



Side section along the strip (thickness exaggerated): clamp base (blue), strip (amber), seal tape (orange)



Also Caught: STEP exported in cm (now mm); a drawing showed 0.13 for 0.125 mm (fixed).

Potential Solutions & Risk Mitigation

FIXED · DESIGNED · FUTURE

Fixed & Verified

Seal: starts at x = 23 mm; 0 clash.

Units: all STEP files in mm.

Checks: 99 CAD checks pass.

Designed, to Prove

v0.2 strip: every sense lead looped.

Contact: silver pads, rounded pins.

Self-test: check resistor → FAULT.

Fallbacks

Wider lines: if E2 says so.

Process: slower passes, log current.

Concept: one-zone strip (score 4.29).

Future Scope & Validation Plan

- **Experiments:** E2 laser, reader bench test, E4 wet tests.
- **Material:** Raman and SEM before any 'graphene' claim.
- **Reliability:** bending, contact cycling, humidity drift.
- **Product:** battery pod with BLE; strips per dressing model.
- **Before patients:** ISO 10993, IEC 60601-1, ethics approval.
- **Not claimed:** sensing performance, graphene, clinical benefit.

Rule We Keep: no performance, graphene or clinical claim until it is measured.

References & Sources

SOURCES WITH READING DEPTH

Numbers match the superscripts on each slide. After each source we state how deeply it was read; nothing is labelled verified beyond that.

1. Milne S.D. et al., A wearable wound moisture sensor as an indicator for wound dressing change, Int. Wound J. 13(6):1309–1314 (2016). doi:10.1111/iwj.12521 · full text read

2. Dressing instructions for use: some foam dressings say 'change when exudate approaches ~2 cm from the pad edge'; Mölnlycke Mepilex Border Flex brochure uses strike-through reaching 3 edges · manufacturer documents, read as excerpts

3. McColl D., Cartlidge B., Connolly P., Real-time monitoring of moisture levels in wound dressings in vitro, Int. J. Surg. 5(5):316–322 (2007). doi:10.1016/j.ijsu.2007.02.008 · abstract

4. US 5,469,145 and US 5,838,240 (wet diaper detector: detachable strip + clip-on detector); US 9,291,589; US 10,928,344 · patent summaries, claims not read

5. Jiang Y. et al., Wireless, closed-loop, smart bandage with integrated sensors and stimulators, Nat. Biotechnol. (2022). doi:10.1038/s41587-022-01528-3 · listing + coverage

6. Dealey C., Cameron J., Arrowsmith M., A study comparing two objective methods of quantifying the production of wound exudate, J. Wound Care 15(4):149–153 (2006) · abstract (methods disagree)

7. Lin J. et al., Laser-induced porous graphene films from commercial polymers, Nat. Commun. 5:5714 (2014). doi:10.1038/ncomms6714 · full text read

8. CO_2 laser tube current vs PWM setting, engineering measurements, softsolder.com (2022) · not peer-reviewed

9. Li Y. et al., Laser-induced graphene in controlled atmospheres: from superhydrophilic to superhydrophobic surfaces, Adv. Mater. 29:1700496 (2017). doi:10.1002/adma.201700496 · abstract

10. EN 13726-1:2002 / EN 13726:2023, Test methods for wound dressings (absorbency; 'Solution A') · catalogue entry; online link not reachable from our network

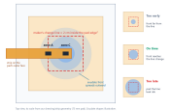
11. DuPont position statement on the use of DuPont materials in medical applications (Kapton caution) · web page

12. Texas Instruments ADS1115 datasheet (SBAS444); Espressif ESP32-C3 datasheet · vendor documents

Image Sources

ALL IMAGES ORIGINAL

Every image in this deck was produced by the team from our own project files. No stock photos, no AI-generated images and no external images are used.



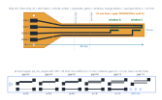
Problem illustration, drawn to scale
Source: make_figures.py · strip_v0_1_geometry.json
1_presentation/figures/fig_problem.png



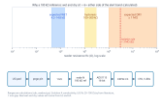
After-state intended workflow
Source: make_figures.py · firmware states
1_presentation/figures/fig_journey_after.png



Firmware scan and state logic
Source: make_figures.py · reader_v0_1 source
1_presentation/figures/fig_dataflow.png



Strip and E2 coupon layers
Source: make_figures.py · our DXF files
1_presentation/figures/fig_strip.png



Measurement window
Source: make_figures.py · calc_readout.py
1_presentation/figures/fig_window.png



Decision matrix + iterations
Source: make_figures.py · decision_matrix.md
1_presentation/figures/fig_decision.png



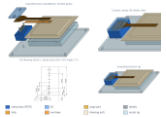
4-month roadmap
Source: make_figures.py · E2/bench/E4 plans
1_presentation/figures/fig_roadmap.png



Before-state pain-point map
Source: make_figures.py · cited sources 1–3
1_presentation/figures/fig_journey_before.png



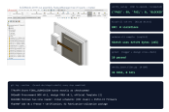
System block diagram
Source: make_figures.py · verification logs
1_presentation/figures/fig_block.png



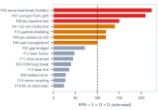
SolidWorks renders + drawing
Source: render_cad.py · 5_cad/solidworks (our models)
1_presentation/figures/fig_cad.png



KiCad schematic export
Source: kicad-cli · reader_v0_1.kicad_sch (ours)
1_presentation/figures/fig_schematic.png



SolidWorks screenshot + logs
Source: screen capture of our machine · verbatim tool output
1_presentation/figures/fig_maker.png



FMEA risk ranking
Source: make_figures.py · fmea_v0_1.csv
1_presentation/figures/fig_fmea.png



Seal clash + lead-break diagrams
Source: make_figures.py · geometry + FMEA
1_presentation/figures/fig_challenges.png